

CSS Margins

Margins are used to create space around elements, outside of any defined borders. The CSS `margin` properties are used to create space around elements, outside of any defined borders.

With CSS, you have full control over the margins. There are properties for setting the margin for each side of an element (top, right, bottom, and left).

Margin - Individual Sides

CSS has properties for specifying the margin for each side of an element:

- `margin-top`
- `margin-right`
- `margin-bottom`
- `margin-left`

All the margin properties can have the following values:

- `auto` - the browser calculates the margin
- `length` - specifies a margin in px, pt, cm, etc.
- `%` - specifies a margin in % of the width of the containing element
- `inherit` - specifies that the margin should be inherited from the parent element

Tip: Negative values are allowed.

Example

Set different margins for all four sides of a `<p>` element:

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  margin-top: 100px;
  margin-bottom: 100px;
  margin-right: 150px;
  margin-left: 80px;
  background-color: lightblue;
}
</style>
</head>
<body>
<h2>Using individual margin properties</h2>
<div>This div element has a top margin of 100px, a right margin of 150px, a bottom margin of 100px, and a left margin of 80px.</div>
</body>
</html>
```

CSS Margins

Margin - Shorthand Property

To shorten the code, it is possible to specify all the margin properties in one property.

The `margin` property is a shorthand property for the following individual margin properties:

- `margin-top`
- `margin-right`
- `margin-bottom`
- `margin-left`

So, here is how it works:

If the `margin` property has four values:

- `margin: 25px 50px 75px 100px;`
 - top margin is 25px
 - right margin is 50px
 - bottom margin is 75px
 - left margin is 100px

Example

Use the margin shorthand property with four values:

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  margin: 25px 50px 75px 100px;
  background-color: lightblue;
}
</style>
</head>
<body>
<h2>The margin shorthand property - 4 values</h2>
<div>This div element has a top margin of 25px, a right margin of 50px, a
bottom margin of 75px, and a left margin of 100px.</div>
<hr>
</body>
</html>
```

CSS Margins

If the `margin` property has three values:

- **`margin: 25px 50px 75px;`**
 - top margin is 25px
 - right and left margins are 50px
 - bottom margin is 75px

Example

Use the margin shorthand property with three values:

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  margin: 25px 50px 75px;
  background-color: lightblue;
}
</style>
</head>
<body>
```

```
<h2>The margin shorthand property - 3 values</h2>
```

```
<div>This div element has a top margin of 25px, a right and left margin
of 50px, and a bottom margin of 75px.</div>
```

```
<hr>
```

```
</body>
</html>
```

CSS Margins

If the `margin` property has two values:

- **margin: 25px 50px;**
 - top and bottom margins are 25px
 - right and left margins are 50px

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  margin: 25px 50px;
  background-color: lightblue;
}
</style>
</head>
<body>
```

```
<h2>The margin shorthand property - 2 values</h2>
```

```
<div>This div element has a top and bottom margin of 25px, and a right
and left margin of 50px.</div>
```

```
<hr>
```

```
</body>
</html>
```

If the `margin` property has one value:

- **margin: 25px;**
 - all four margins are 25px

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  margin: 25px;
  background-color: lightblue;
}
</style>
</head>
<body>
<h2>The margin shorthand property - 1 value</h2>
<div>This div element has a top, bottom, left, and right margin of 25px.</div>
<hr>
</body>
</html>
```

CSS Margins

The auto Value

It can set the margin property to **auto** to horizontally center the element within its container.

The element will then take up the specified width, and the remaining space will be split equally between the left and right margins.

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width:300px;
  margin: auto;
  border: 1px solid red;
}
</style>
</head>
<body>
<h2>Use of margin:auto</h2>
<p>You can set the margin property to auto to horizontally center the element within its container. The element will then take up the specified width, and the remaining space will be split equally between the left and right margins:</p>
<div>
This div will be horizontally centered because it has margin: auto;
</div>
</body>
</html>
```

The inherit Value

This example lets the left margin of the `<p class="ex1">` element be inherited from the parent element (`<div>`):

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid red;
  margin-left: 100px;
}

p.ex1 {
  margin-left: inherit;
}
</style>
</head>
<body>
<h2>Use of the inherit value</h2>
<p>Let the left margin be inherited from the parent element:</p>
<div>
<p class="ex1">This paragraph has an inherited left margin (from the div element).</p>
</div>
</body>
</html>
```

CSS Margins

CSS Margin Collapse

Sometimes two margins collapse into a single margin.

Margin Collapse

Top and bottom margins of elements are sometimes collapsed into a single margin that is equal to the largest of the two margins.

This does not happen on left and right margins! Only top and bottom margins!

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  margin: 0 0 50px 0;
}

h2 {
  margin: 20px 0 0 0;
}
</style>
</head>
<body>
```

`<p>`In this example the h1 element has a bottom margin of 50px and the h2 element has a top margin of 20px. So, the vertical margin between h1 and h2 should have been 70px (50px + 20px). However, due to margin collapse, the actual margin ends up being 50px.`</p>`

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>

</body>
</html>
```

All CSS Margin Properties

Property	Description
<u>margin</u>	A shorthand property for setting the margin properties in one declaration
<u>margin-bottom</u>	Sets the bottom margin of an element
<u>margin-left</u>	Sets the left margin of an element
<u>margin-right</u>	Sets the right margin of an element
<u>margin-top</u>	Sets the top margin of an element